ARUN M
Email:Aruntrijal7@gmail.com **Mobile :6304592628**

PROFILE:

- 4+ years of experience of wireless telecom domain in 5g NR gNodeB protocol development and framework development.
- Strong system design and architecture definition capabilities
- Work experience on the LTE/5G NR Radio L3/L2 protocols.
- Good exposure to MAC and RLC.
- Having good knowledge on RTOS.
- Having good knowledge on Python Automation scripting .
- Deep insight of Linux operating systems with knowledge on multi-threaded programming.
- Worked with LTE eNodeB,5g NR SA, LTE-Layer3 (RRC) on product gnodeb.
- Excellent understanding and deep insight in 3GPP 4G and 5G specification (RAN).
- Having good knowledge on LTE & 5G NR architecture, functionalities, basic call flow and Handover Call flows.
- Extensively involved in complete software development life cycle including planning, analysis, design and implementation and testing.
- Involved in Customer, Sprint demos, retros.
- Experienced in cross-functional teams, involved in both development and testing.
- Having good Understanding and knowledge on Standards like 3GPP, LTE and NR.
- Good exposure to C, C++ and Data Structures on Linux Environment including debugging capabilities.
- Handling and debugging of customer issues, reproduction of failed scenarios to resolve customer blocking issues and support product commercialization.
- Excellent technical and functional debugging and problem-solving capabilities and easy grasping of new things.

EXPERTISE:

- Programming Languages : C, C++ and Data Structures , Python.
- Operating Systems : Linux, Windows
- Technology : LTE, 5G NR, SA
- Protocols : RRC, X2AP, RLC/MAC
- Protocol Analyzer : Wireshark
- CI/CD : GIT, Jenkins
- Issue Tracking : JIRA

EDUCATION:

- B.Tech from Jawaharlal Nehru Technological University in 2017.

EMPLOYMENT:

- Worked in **Mintap Techno Service LLP** since 03-06-2020

PROFESSIONAL EXPERIENCE:

2022-07 to 2025-03 Software Engineer.

Project: 5g SA NR Cell Resource Co-ordination.

Description:

In 5G Standalone (SA) NR networks, **Cell Resource Coordination** involves the exchange of signaling messages between **gNBs** (over **Xn interface**) or between **gNB and NG-RAN functions** to **optimize resource usage, manage interference, and improve mobility**.

Responsibilities:

- Implement **Cell Resource Coordination Request & Response** message handling as per **3GPP NR specifications**.
- Develop logic for:
- Resource status reporting (available PRBs, load level, interference level)
- Neighbor cell capability exchange
- Dynamic spectrum coordination
- Integrate **XnAP message handling** with **gNB RRC**

2021-04 to 2022-06 System Engineer.

Project: 3GPP LTE RAN- RRC Protocol Customer support.

Description:

This project involves **customer-facing technical support** for LTE Radio Access Network (RAN) software, focusing specifically on the **RRC protocol (Layer 3)** per 3GPP TS 36.331. The goal is to help operators, integrators, and OEM partners **deploy, optimize, and troubleshoot** LTE networks by providing deep technical expertise on **RRC signaling flows, mobility procedures, and interoperability**.

Responsibilities:

Analyze **RRC signaling logs** (using Wireshark, QXDM, TEMS) for:

- RRC connection failures
- Handover failures
- Radio Link Failures (RLF)
- Attach/Detach issues
- Paging problems

2020-07 to 2021-03 System Engineer.

Project: eNodeB RRC/L3 Development.

Description:

The project involved development of a protocol flow for eNB in LTE environment. The **eNodeB RRC/L3 Developer** is responsible for designing, implementing, and maintaining the Radio Resource Control (RRC) and Layer 3 protocols of an LTE eNodeB (Evolved NodeB). This role involves working on **3GPP-compliant signaling procedures**, ensuring **seamless UE-network communication**, optimizing performance, and debugging protocol-level issues.

- Implement **RRC layer** functionalities as per **3GPP TS 36.331** specifications.
- Develop and maintain **Layer 3 call processing modules** for eNodeB.
- Implement key **RRC procedures** such as:
- **RRC Connection Establishment / Release**
- **RRC Reconfiguration**
- **Handover (Intra-LTE, Inter-RAT)**
- **Measurement control & reporting**
- **Paging handling**
- **Cell broadcast messages (SIBs, MIBs)**

(Arun M)